Drum Classification

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Outline:

Motivation

Dataset

Feature Engineering

Models

Results

Motivation

- Love music, stepping-stone to more interesting applications
- Personal Dataset

Dataset

Personal Splice Library

Drums:

- *Kicks*
- *Snares*
- *Hats*
- *Rides*

Extracted through regex patterns in pc files

- Messy data, requires filtering

Insert Audio Sample Play Button

Feature Engineering

Max Amplitude

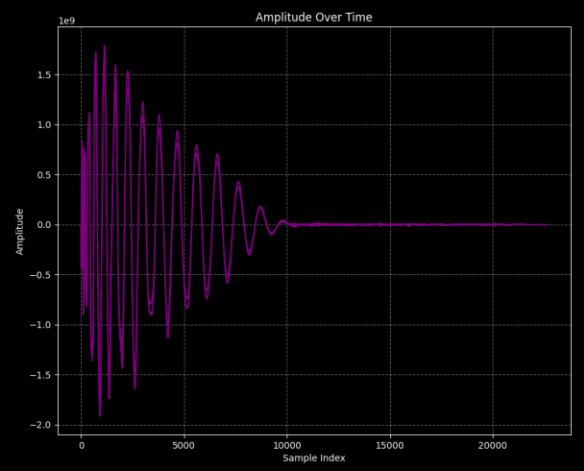
Highest level of loudness

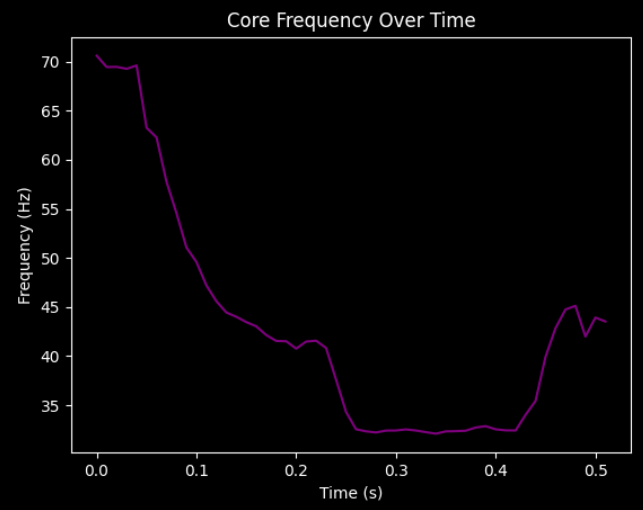
Core Frequency

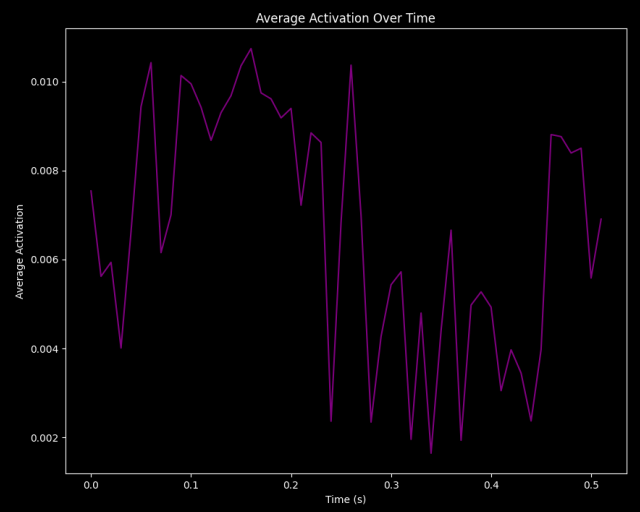
$$\hat{c} = \frac{\sum_{i=1}^{360} \hat{y}ic_i}{\sum i = 1^{360} \hat{y}i}, \quad \hat{f} = f\mathrm{ref} \cdot 2^{\hat{c}/1200}$$

Average Activation

Confidence of pitch (???)







Extracted using Crepe Library

Data Cleaning

Filtered Out: Perc, Clap, Tom

Merged: Crash into ride

Balanced: To min snare

Initial

label	
kick	471
snare	467
tom	250
clap	138
ride	120
hat	111

Filtered

label	
kick	467
snare	461
ride	111
hat	103

Balanced

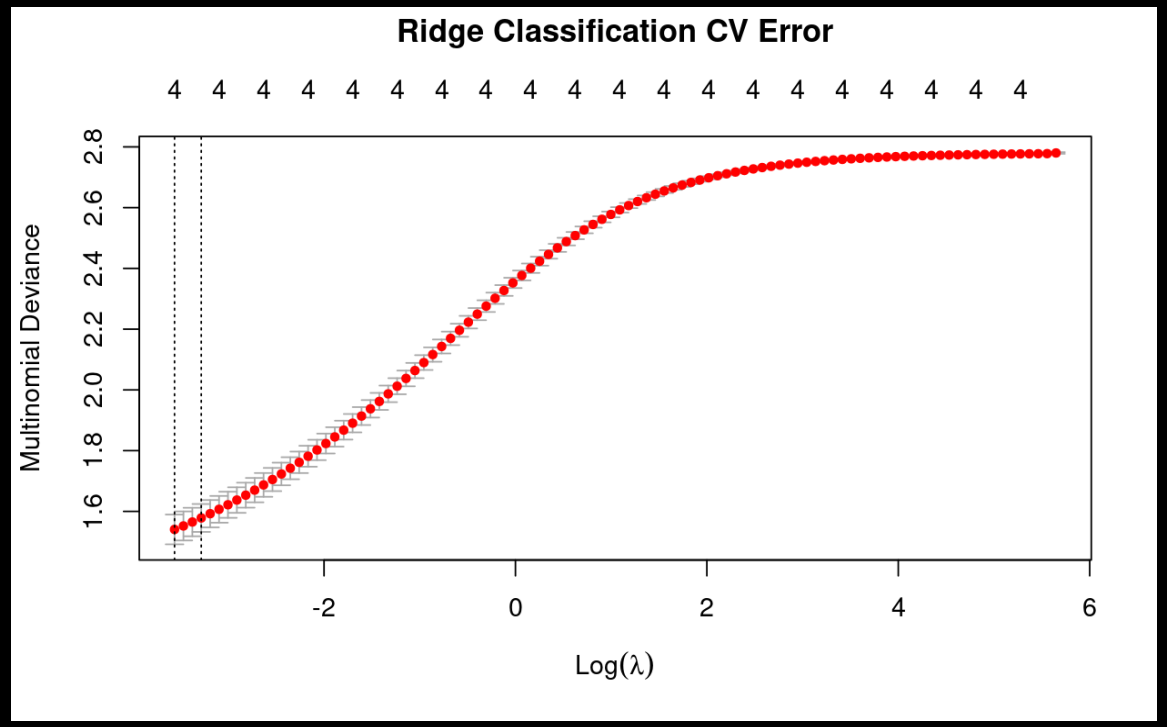
label	
hat	103
kick	103
ride	103
snare	103

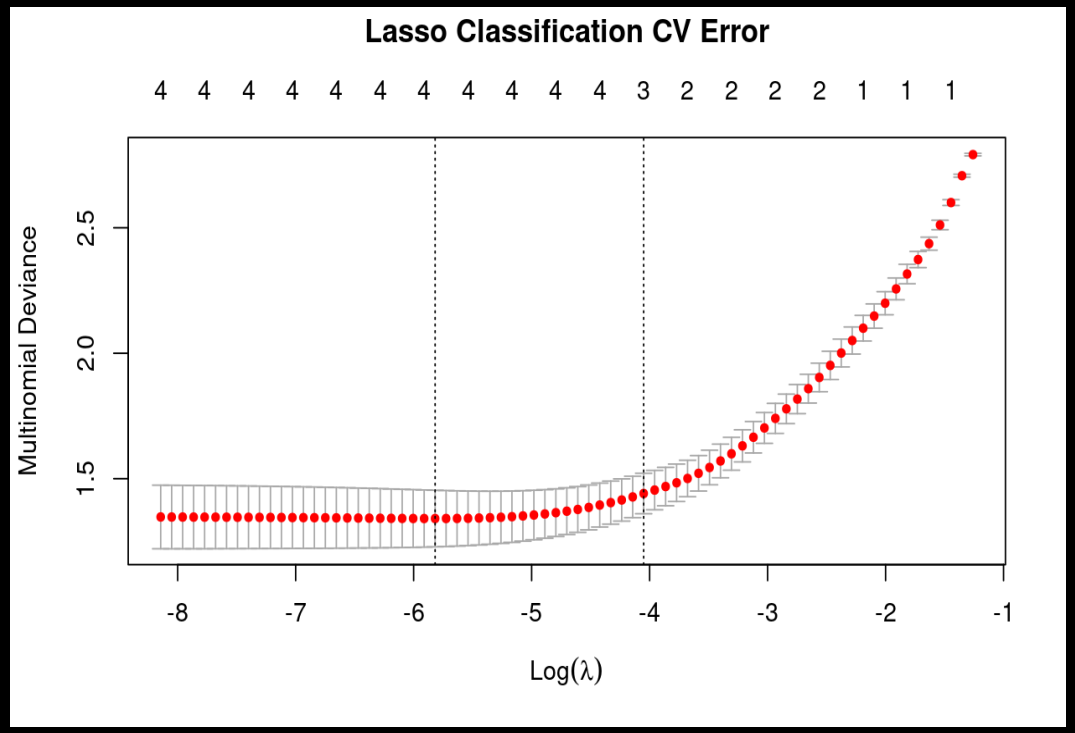
Models

Ridge and Lasso
SVM
Tree

Classification CV Error

- Ridge shows a smoother curve





- Lasso is better at ignoring less important features
- Better handles feature selection

Classification CV Error

- Ridge:
  - Error minimized around $log(\lambda) = -2$ the error rises after due to overfitting
  - Smaller lambda introduces higher variance, leads to overfitting

- Lasso:
  - Error minimized around $log(\lambda) = -6$, and remains stable across wider ranges compared to Ridge

Accuracy

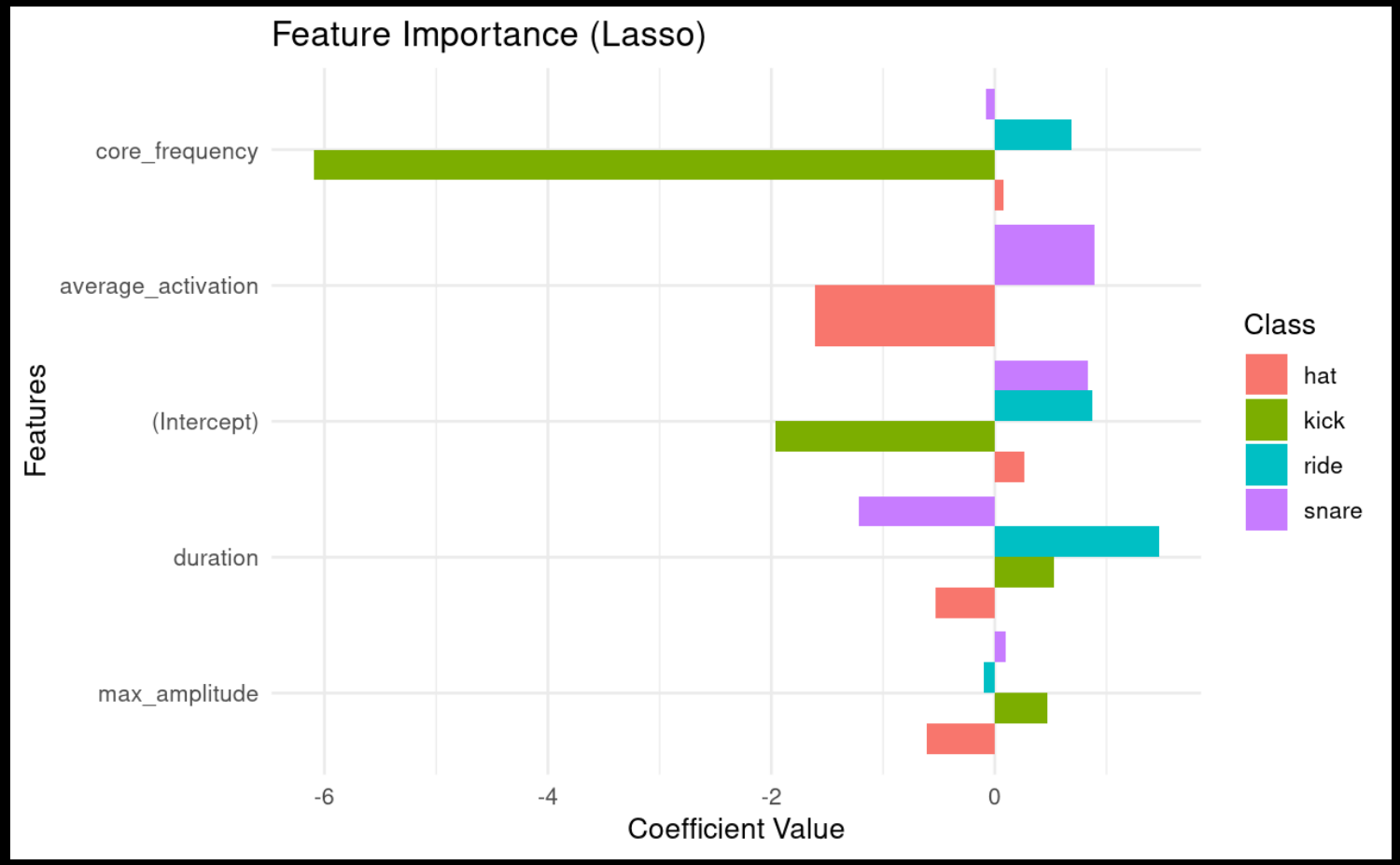
- Lasso preforms the best among the three models achieving 82.5% accuracy

- Tuning does not seem to improve Ridge or Lasso

```
[1] "Lasso Accuracy: 82.5 %"
[1] "Ridge Accuracy: 80 %"
[1] "Tuned Lasso Accuracy: 82.5 %"
[1] "Tuned Ridge Accuracy: 80 %"
[1] "Elastic Net Accuracy: 81.25 %"
```

Feature Importance (Lasso)

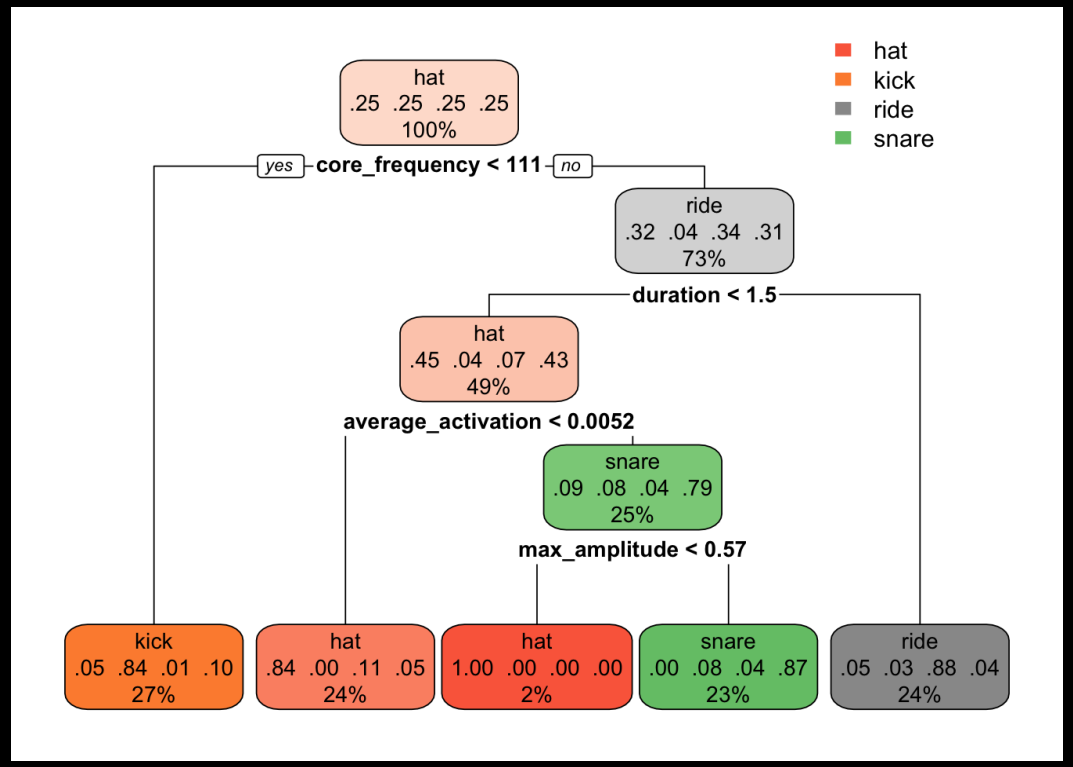
- Core frequency: significant for "kick" and "ride"
- Average activation: helps differentiate "snare" and "hat'
- Other Features: duration and max amplitude are less significant
- Lasso simplified the feature set, improving interpretability



Tree Model

Initial

- 75% Training, 25% Testing
  - Stratified by 'label'
- decision_tree() function
- Five terminal nodes
- Four Decisions
  - Core_freq < 111
  - Duration < 1.5
  - Avg_activation < 0.0052
  - Max_amplitude < 0.57

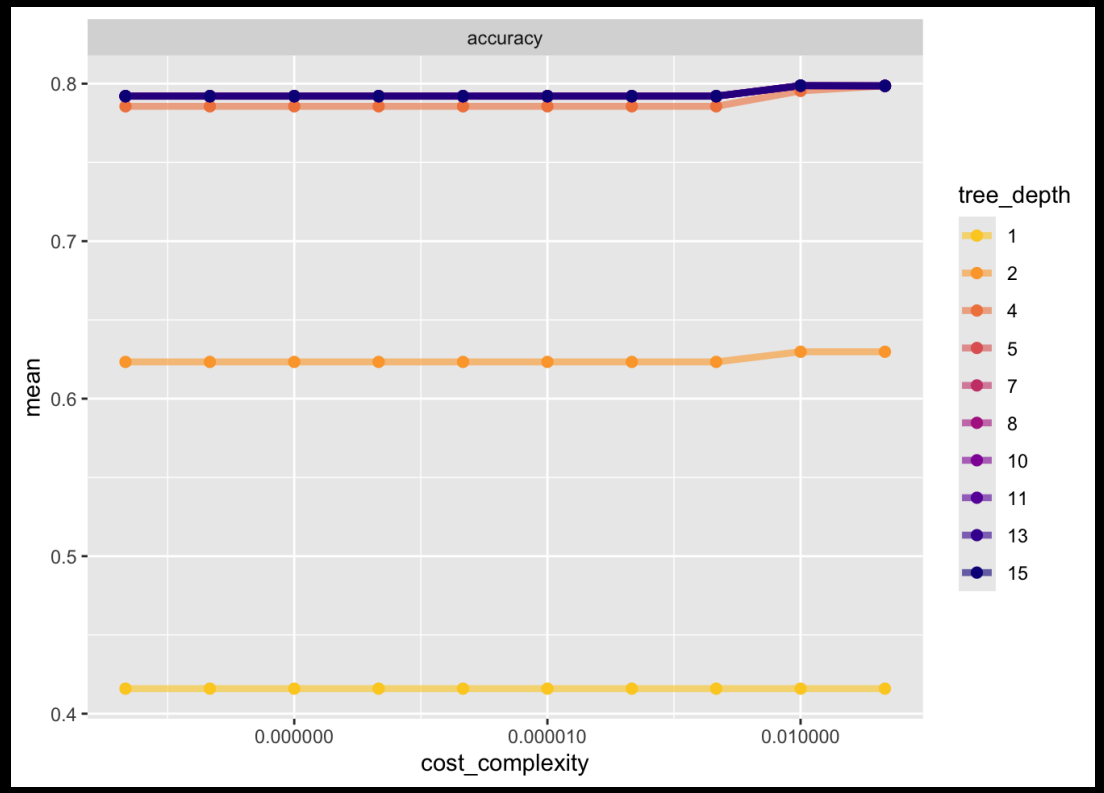


Tree Model

Tuning

- 10-fold cross-validation used to find best values for cost complexity

- Highest accuracy with minimal complexity:
  - Cost complexity = 0.01

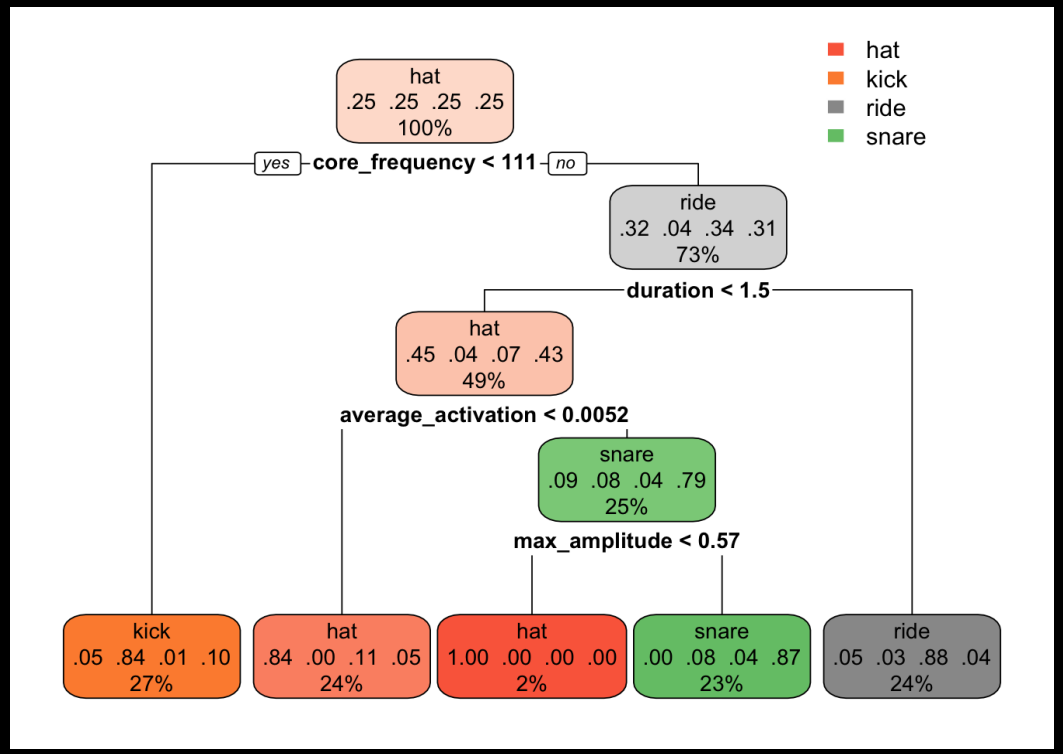


Tree Model

Final

- Final model with a tuned cost complexity parameter
- Overall Test Accuracy: 0.865

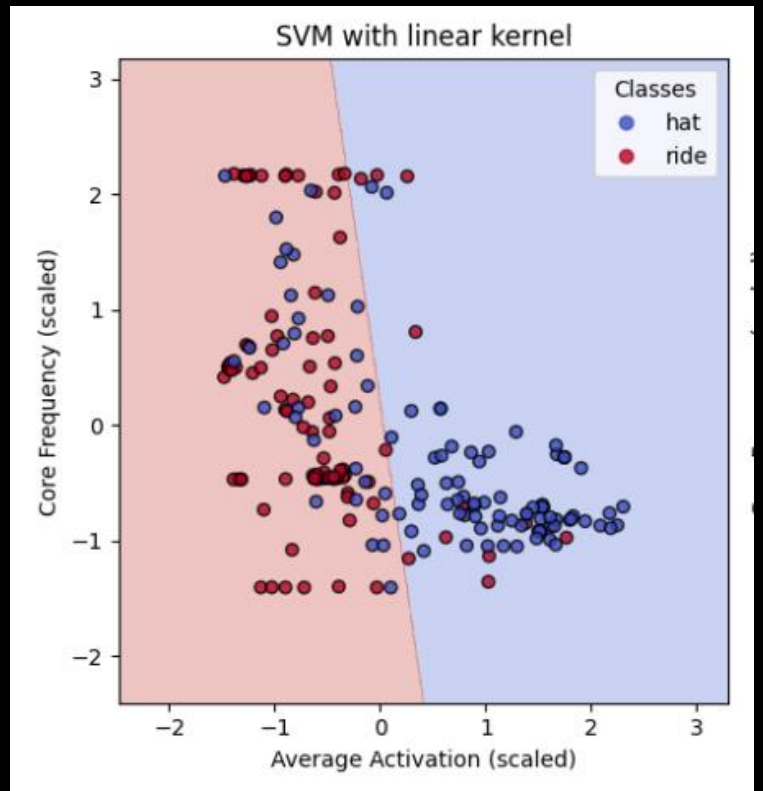
```
##           Reference
## Prediction hat kick ride snare
##      hat    22    3    1     0
##      kick    0   25    1     0
##      ride    1    0   25     0
##      snare   3    4    1    18
```

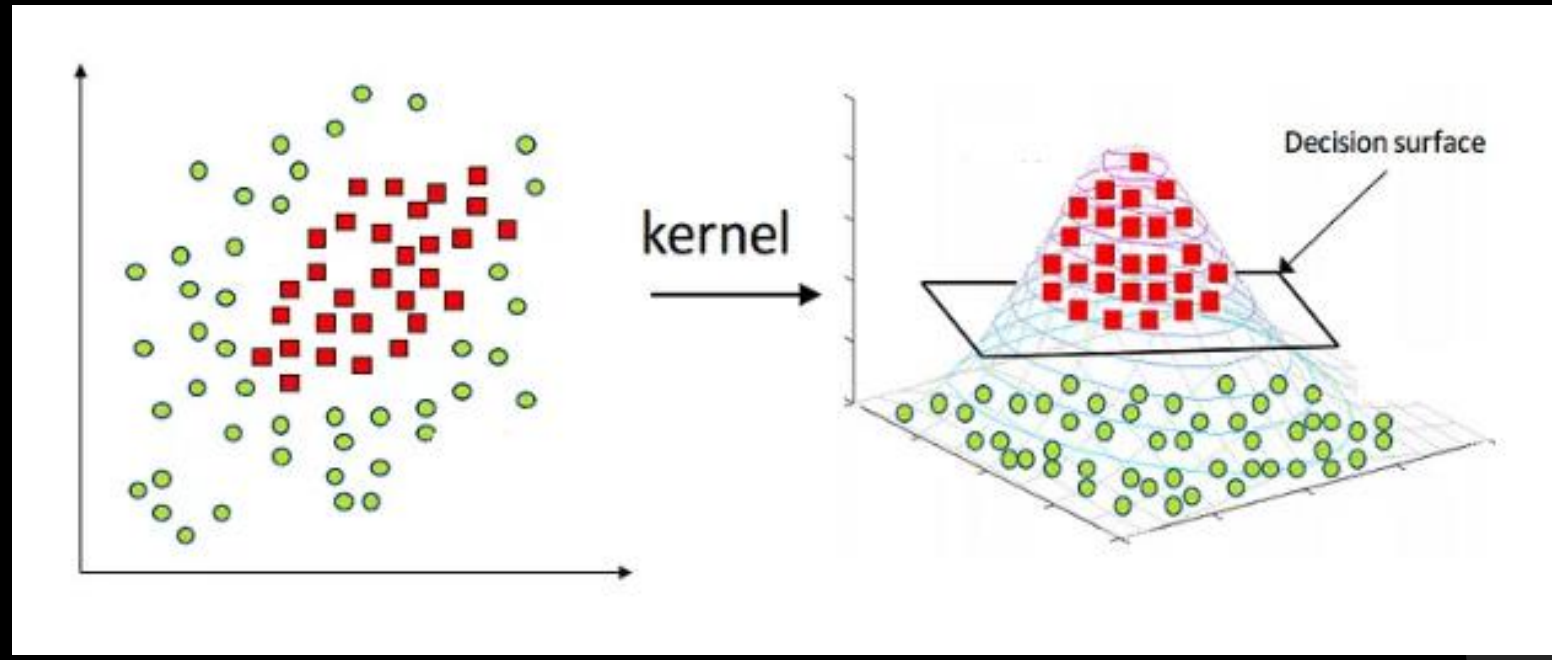


SVM

What is SVM?

- SVC vs SVR
- Finding support vectors
- Kernels

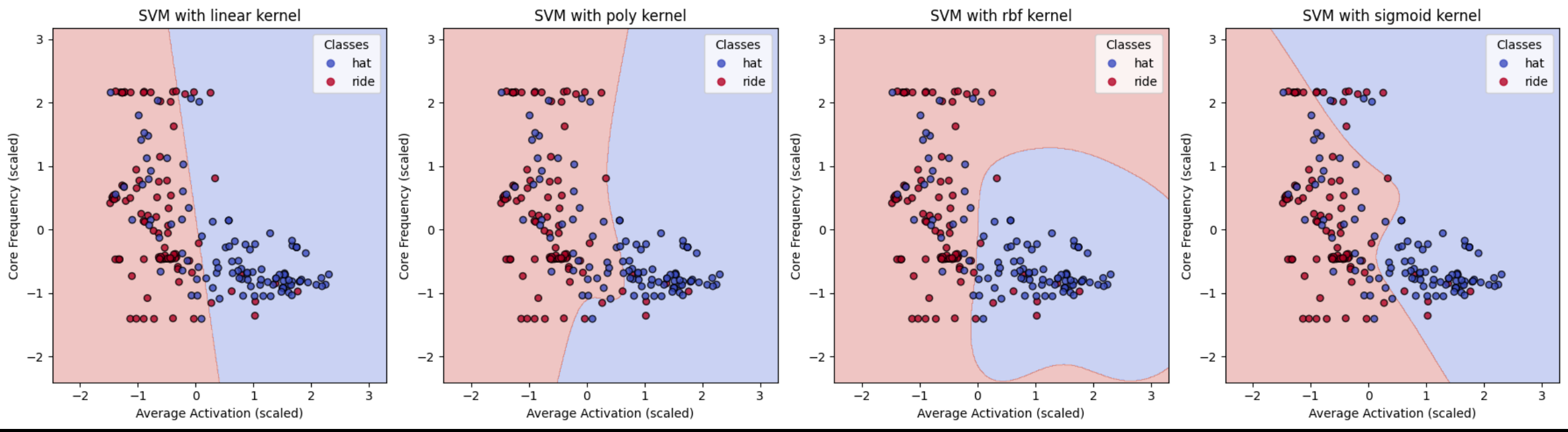




Source: https://medium.com/@Suraj_Yadav/what-is-kernel-trick-in-svm-interview-questions-related-to-kernel-trick-97674401c48d

SVM

Exploring Kernels



	Linear	Poly	RBF	Sigmoid	Average
Binary w/ 2 features	0.774194	0.741935	0.758065	0.709677	0.745968

SVM

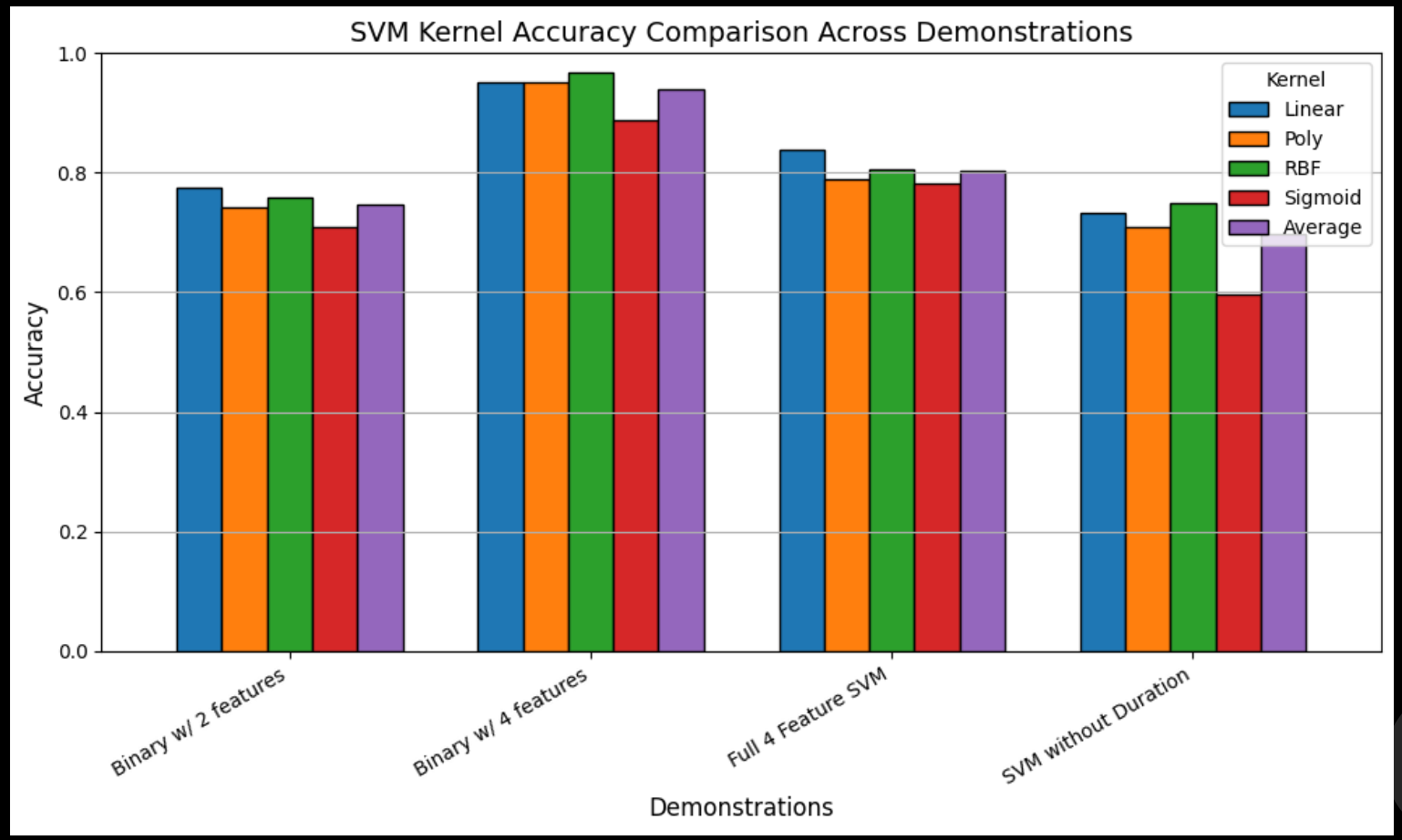
Using the whole dataset

- 10-fold CV
- Precision, Recall, and f1-score
- Linear kernel struggled with hat
- Linear kernel excelled with snare
- Inverse results for rest of kernels
- Linear:

	precision	recall	f1-score
hat	0.66	0.95	0.78
kick	0.86	0.91	0.89
ride	0.88	0.76	0.81
snare	0.96	0.78	0.86

- Linear:

	precision	recall	f1-score
hat	0.89	0.89	0.89
kick	0.75	0.77	0.76
ride	0.93	0.84	0.88
snare	0.63	0.68	0.65



Results

- Ridge/Lasso best accuracy is 82.5%
- Decision tree best accuracy is 85.6%
- SVM best accuracy with full data is 83.8% with linear kernel.